SIMATS ENGINEERING
CO5-ASSESSMENT TOOL -14
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Course Code: **CSA1511** **Slot:** B

Course Name: **CLOUD COMPUTING AND BIG DATA ANALYTICS**

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1. Difference between HDFS and traditional file systems:

HDFS stores large files across multiple computers, while traditional file systems usually store files on a single computer or server.

2. Why is replication important in distributed storage?

Replication keeps multiple copies of data. If one computer fails, the data can still be accessed from another computer.

3. How does MapReduce divide work between Map and Reduce phases?

- **Map:** Processes and organizes input data into key-value pairs.
- **Reduce:** Combines and summarizes the results from the Map phase.

4. What role does YARN play in Hadoop?

YARN manages system resources and schedules tasks across the Hadoop cluster.

5. Difference between NAS and SAN:

- **NAS:** Provides file-level storage through a network.
- **SAN:** Provides block-level storage and is generally used for high-performance enterprise systems.

6. What is the purpose of ETL in a big data pipeline?

ETL means **Extract, Transform, and Load**. It collects data, cleans or transforms it, and stores it for analysis.

7. How does Apache NiFi help in data integration?

Apache NiFi helps collect, move, transform, and manage data between different systems automatically.

8. What are key failure points in a Hadoop cluster and how would you handle them?

Common failures include **NameNode failure, DataNode failure, and network failure**. They can be handled using replication, backups, and high-availability systems.

9. How would you design a secure data ingestion pipeline?

Use authentication, encryption, access control, secure connections, and regular monitoring to protect the data.

10. How would you explain a scalable Hadoop system to a non-technical manager?

A Hadoop system can handle very large amounts of data by dividing the work among many computers. More computers can be added when the data grows.